

Hembree_Alex_Assignment9&10

May 10, 2026

0.0.1 DSC 640 - Data Presentation & Visualization

0.0.2 Week 9 & 10, Assignment - TSA Complaints

0.0.3 Hembree, Lex

For this exercise you will be analyzing TSA complaint data. There are 4 datasets below. 3 provide information on the complaints, counts by Airport, counts by category, and the final dataset includes the airport name by code. For this exercise, you must craft a story about the data based on the following parameters:

Audience: As you craft your story, you need to determine who your audience will be and the level of data understanding they possess.

- Are they familiar with the data, potentially an employee or stakeholder? Or are they someone unfamiliar with the data, like the general public? - This distinction is important because it will determine how much detail you provide and what kind of jargon you can use with the audience.
- In addition to the audience's familiarity with the data, consider their level of knowledge. Are they executives, board members, employees, children, etc.? This is an important piece of information to keep in mind as it will tailor the visuals you use and again how you present the information and what your audience might be interested in. Consider the scenario where you are creating a story for a parent vs the kids and how different your messaging would be based on those two different audiences.

Purpose:

- The story you are presenting must include a call to action. It cannot simply be an informational presentation.
- What is the intent of your message/story? What do you want to get across?
- What are you hoping your audience walks away knowing and understanding?
- Are you asking for approval, recommending a change, or promoting a new direction?

Medium:

- How will you present the information?
- Will you create a PowerPoint presentation, dashboard, infographic, video, TikTok, Reel, billboard?
- What is the best method to convey the information based on your selected audience?

Design:

- How did you consider Gestalt's principles in your design decisions?
- What was your strategy with the following elements?
 - Color
 - Text
 - Alignment
 - Sizing
 - Spacing

Visuals:

- 3 of your 6 required visualizations must be from this list of types of visual:
 - Heat Maps
 - Custom Visuals
 - Box Plots
 - Spatial Visuals/Choropleth Maps

Ethical Considerations:

- What changes were made to the data?
- Are there any legal or regulatory guidelines for your data?
- What risks could be created based on the transformations or how the visualizations are presented?
- Did you make any assumptions in cleaning/transforming or when presenting the data? Did you filter any data without labeling or clearly identifying that the data was not included?
- How was your data sourced/verified for credibility?
- Was your data acquired in an ethical way?
- How would you mitigate any of the ethical implications you have identified?

To get started, analyze the data and determine what looks interesting. You are welcome to make things up, role play, have fun with these exercises.

Datasets:

- Complaints by Airport
- Complaints by Category
- Complaints by Subcategory
- IATA/ICAO

Dataset Sources:

- Provides the specific complaint information across 3 files: GitHub tsa-complaint-counts
- Provides the airport location & look-up codes, which will be needed to narrow in on what airport had specific TSA complaints: IATA/ICAO

Requirements:

- You must use the 4 datasets provided for whatever medium you are choosing to present your story and remember it must include a call to action.

- This exercise can use PowerBI/Tableau/R or Python. Make sure to carefully check each exercise, as some exercises will require a specific tool be used.
- You must define an audience, purpose, and medium to tell your story.
- You must have at least 6 visuals included that use the data provided and 3 must be from this list:
 - Heat Maps
 - Custom Visuals
 - Box Plots
 - Spatial Visuals / Choropleth Maps
- Include a 250-500-word paper summarizing your audience, purpose, medium and design choices along with the above ethical consideration questions.

FAQs:

- Can I go to the dataset sources to access the most current data?
 - Yes, as long as you include all the data.
- Can I include additional data?
 - Yes, as long as you include the provided 4 datasets.
- How do I include a call to action if I am creating a dashboard, an infographic or a blog post, etc.?
 - With an infographic, you should be able to make it clear you want people to do a certain thing, same with a blog, but agree it could be more challenging to do with a dashboard – this is where I would include what your plan is in your written paper so that it supports the visual story you have created since it wouldn't make sense to include the written narrative on a dashboard like you would on a presentation or blog.
- Can I only include visuals from the included list?
 - Yes.
- If I include 2 line charts, do I still need 3 other types of visuals or does that count toward the 3 of the 6?
 - I want you to get practice using the different tools to create different visuals, so you will need to still practice doing a bar, stacked bar, and line chart for example to meet the requirement. Even if you have duplicates of the same chart, it would not be counted as a separate chart to meet the requirement of having variety from the list.

Submission Instructions:

- Submit your results as one consolidated document (PDF/PowerPoint/Word) via the assignment link. You must also submit your code and any unique data sources you used. Do NOT submit zip files.
- Make sure to carefully review the rubric, as it will be used to grade each exercise.
- The assignment is due by Sunday of Week 10, 11:59 p.m. CT.

```
[1]: # import libraries
import numpy as np
import pandas as pd

# import file as pandas data frame
```

```
airport_df = pd.read_csv("C:/Users/Ashem/OneDrive/Desktop/Masters Program - DSC/
↳DSC 640 - Data Presentation and Visualization/Week 9&10/
↳complaints-by-airport.csv")
category_df = pd.read_csv("C:/Users/Ashem/OneDrive/Desktop/Masters Program -
↳DSC/DSC 640 - Data Presentation and Visualization/Week 9&10/
↳complaints-by-category.csv")
subcategory_df = pd.read_csv("C:/Users/Ashem/OneDrive/Desktop/Masters Program -
↳DSC/DSC 640 - Data Presentation and Visualization/Week 9&10/
↳complaints-by-subcategory.csv")
iata_df = pd.read_csv("C:/Users/Ashem/OneDrive/Desktop/Masters Program - DSC/
↳DSC 640 - Data Presentation and Visualization/Week 9&10/iata-icao.csv")
```

```
[2]: # Preview
airport_df.head()
```

```
[2]: pdf_report_date airport year_month count
0      2019-02      ABE      2015-01      0
1      2019-02      ABE      2015-02      0
2      2019-02      ABE      2015-03      0
3      2019-02      ABE      2015-04      0
4      2019-02      ABE      2015-05      2
```

```
[3]: # Preview
category_df.head()
```

```
[3]: pdf_report_date airport category year_month \
0      2019-02      ABE      Hazardous Materials Safety      2015-01
1      2019-02      ABE      Mishandling of Passenger Property      2015-01
2      2019-02      ABE      Hazardous Materials Safety      2015-02
3      2019-02      ABE      Mishandling of Passenger Property      2015-02
4      2019-02      ABE      Hazardous Materials Safety      2015-03

count clean_cat clean_cat_status
0      0      Hazardous Materials Safety      original
1      0      Mishandling of Passenger Property      original
2      0      Hazardous Materials Safety      original
3      0      Mishandling of Passenger Property      original
4      0      Hazardous Materials Safety      original
```

```
[4]: # Preview
subcategory_df.head()
```

```
[4]: pdf_report_date airport category \
0      2019-02      ABE      Hazardous Materials Safety
1      2019-02      ABE      Mishandling of Passenger Property
2      2019-02      ABE      Hazardous Materials Safety
3      2019-02      ABE      Mishandling of Passenger Property
```

```
4          2019-02      ABE      Hazardous Materials Safety
```

```

                                subcategory year_month  count  \
0                                General    2015-01        0
1  Damaged/Missing Items--Checked Baggage  2015-01        0
2                                General    2015-02        0
3  Damaged/Missing Items--Checked Baggage  2015-02        0
4                                General    2015-03        0

```

```

                                clean_cat                                clean_subcat  \
0      Hazardous Materials Safety                                General
1  Mishandling of Passenger Property  *Damaged/Missing Items--Checked Baggage
2      Hazardous Materials Safety                                General
3  Mishandling of Passenger Property  *Damaged/Missing Items--Checked Baggage
4      Hazardous Materials Safety                                General

```

```

clean_cat_status clean_subcat_status  is_category_prefix_removed
0      original          original                False
1      original          original                False
2      original          original                False
3      original          original                False
4      original          original                False

```

```
[5]: # Preview
iata_df.head()
```

```

[5]:  country_code region_name iata  icao      airport  \
0      AE      Abu Zaby  AAN  OMAL      Al Ain International Airport
1      AE      Abu Zaby  AUH  OMAA  Abu Dhabi International Airport
2      AE      Abu Zaby  AYM   NaN      Yas Island Seaplane Base
3      AE      Abu Zaby  AZI  OMAD      Al Bateen Executive Airport
4      AE      Abu Zaby  DHF  OMAM      Al Dhafra Air Base

```

```

latitude longitude
0  24.2617    55.6092
1  24.4330    54.6511
2  24.4670    54.6103
3  24.4283    54.4581
4  24.2482    54.5477

```

```
[6]: list(airport_df.columns)
```

```
[6]: ['pdf_report_date', 'airport', 'year_month', 'count']
```

```
[7]: list(category_df.columns)
```

```
[7]: ['pdf_report_date',  
      'airport',  
      'category',  
      'year_month',  
      'count',  
      'clean_cat',  
      'clean_cat_status']
```

```
[8]: list(subcategory_df.columns)
```

```
[8]: ['pdf_report_date',  
      'airport',  
      'category',  
      'subcategory',  
      'year_month',  
      'count',  
      'clean_cat',  
      'clean_subcat',  
      'clean_cat_status',  
      'clean_subcat_status',  
      'is_category_prefix_removed']
```

```
[9]: list(iata_df.columns)
```

```
[9]: ['country_code',  
      'region_name',  
      'iata',  
      'icao',  
      'airport',  
      'latitude',  
      'longitude']
```

Visual 1 — Heat Map (complaints by airport × month)

```
[10]: # import libraries  
import matplotlib.pyplot as plt  
import matplotlib  
import seaborn as sns  
  
# define  
heat_df = (  
    category_df.groupby(["category", "year_month"])["count"]  
    .sum()  
    .reset_index()  
    .pivot(index="category", columns="year_month", values="count")  
    .fillna(0)  
)
```

```

# sort
cat_order = (
    heat_df.sum(axis=1)
    .sort_values(ascending=False)
    .index
)
heat_df_sorted = heat_df.loc[cat_order]

# compute percentile
vmax_cap = heat_df_sorted[heat_df_sorted > 0].quantile(0.95).max()

# plot
plt.figure(figsize=(20, 16))
ax = sns.heatmap(
    heat_df_sorted,
    cmap="Reds",
    linewidths=0.3,
    norm=matplotlib.colors.LogNorm(vmin=1, vmax=vmax_cap)
)

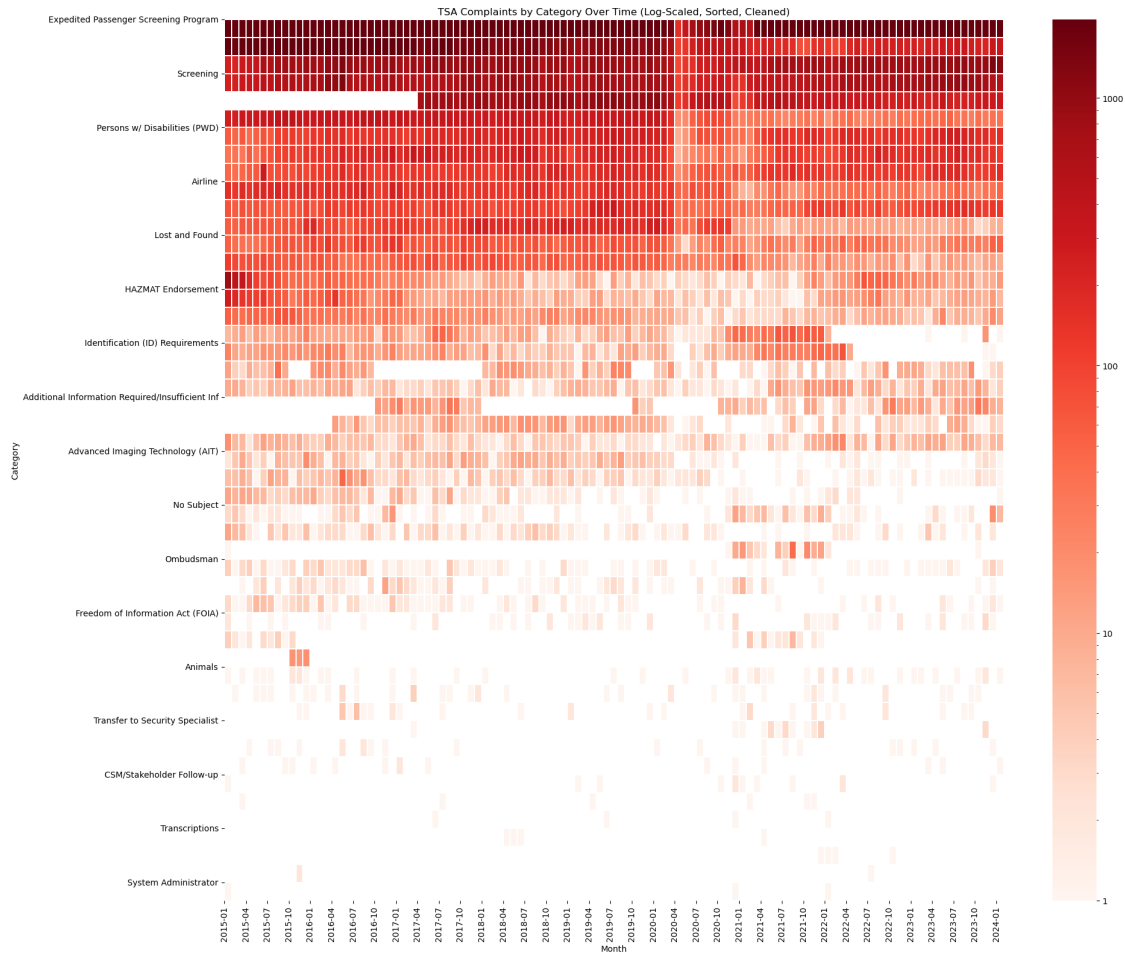
# remove scientific notation from colorbar
ax.collections[0].colorbar.formatter = matplotlib.ticker.
    ↪FormatStrFormatter('%d')
ax.collections[0].colorbar.update_ticks()

# thin y-axis labels
plt.yticks(
    ticks=range(0, len(heat_df_sorted.index), 3),
    labels=heat_df_sorted.index[:3]
)

# thin x-axis labels
plt.xticks(
    ticks=range(0, len(heat_df_sorted.columns), 3),
    labels=heat_df_sorted.columns[:3],
    rotation=90
)

plt.title("TSA Complaints by Category Over Time (Log-Scaled, Sorted, Cleaned)")
plt.xlabel("Month")
plt.ylabel("Category")
plt.tight_layout()
plt.show()

```



Visual 2 — Box Plot (distribution by category)

```
[11]: # define
monthly = (
    category_df.groupby(["category", "year_month"])["count"]
    .sum()
    .reset_index()
)

# shorten labels
monthly["short_cat"] = monthly["category"].apply(
    lambda x: x[:30] + " ..." if len(x) > 30 else x
)

# Plot
plt.figure(figsize=(18, 10))
sns.boxplot(
    data=monthly,
```



```

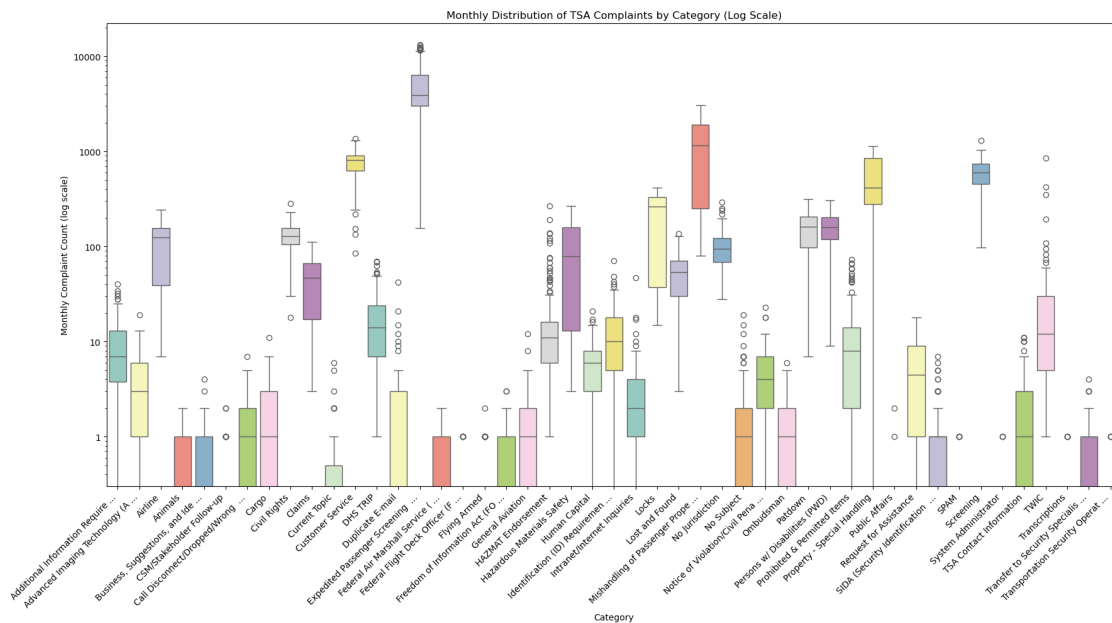
x="short_cat",
y="count",
hue="short_cat",
palette="Set3",
legend=False
)

plt.yscale("log")

# remove scientific notation on primary y-axis
plt.gca().yaxis.set_major_formatter(matplotlib.ticker.ScalarFormatter())
plt.gca().ticklabel_format(style='plain', axis='y')

plt.xticks(rotation=45, ha="right")
plt.title("Monthly Distribution of TSA Complaints by Category (Log Scale)")
plt.xlabel("Category")
plt.ylabel("Monthly Complaint Count (log scale)")
plt.tight_layout()
plt.show()

```



Visual 3 — Spatial Visual (airport complaints on a map)

```

[12]: # filter to us
iata_us = iata_df[iata_df["country_code"] == "US"]

# define
airport_totals = (

```

```

    airport_df.groupby("airport")["count"]
        .sum()
        .reset_index()
)

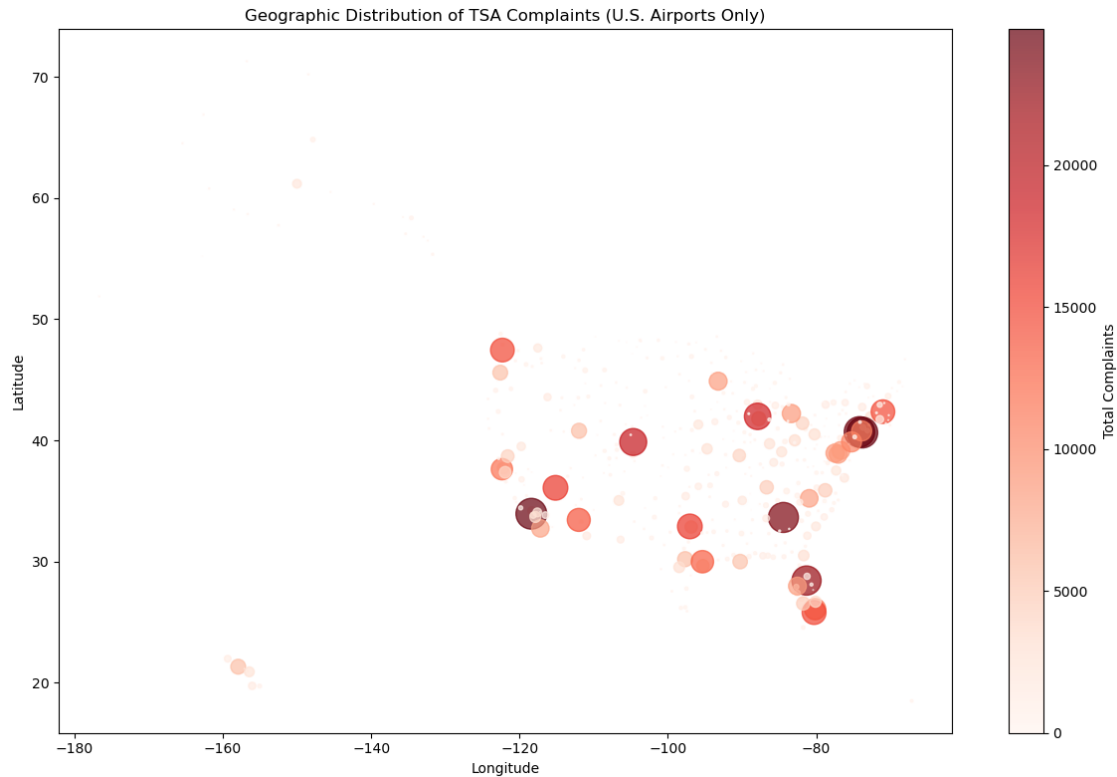
# Merge with IATA coordinates
map_df = airport_totals.merge(
    iata_us,
    left_on="airport",
    right_on="iata",
    how="left"
)

# drop na's
map_df = map_df.dropna(subset=["latitude", "longitude"])

# plot
plt.figure(figsize=(12, 8))
plt.scatter(
    map_df["longitude"],
    map_df["latitude"],
    s=map_df["count"] / 50,
    c=map_df["count"],
    cmap="Reds",
    alpha=0.7
)

plt.colorbar(label="Total Complaints")
plt.title("Geographic Distribution of TSA Complaints (U.S. Airports Only)")
plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.tight_layout()
plt.show()

```



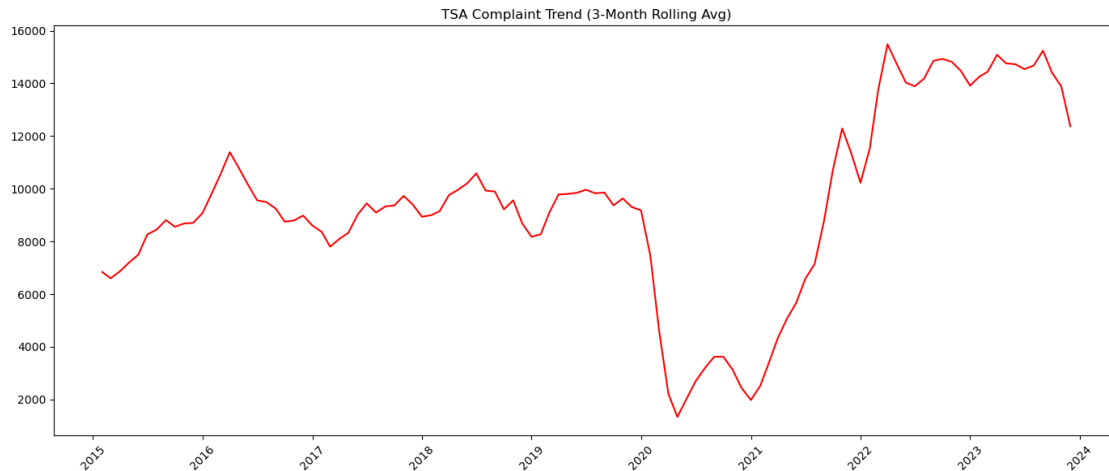
Visual 4 — Time Series Trend (total complaints per month)

```
[13]: # define
airport_df["year_month"] = pd.to_datetime(airport_df["year_month"])

trend_df = (
    airport_df.groupby("year_month")["count"]
    .sum()
    .reset_index()
    .sort_values("year_month")
)

trend_df["rolling"] = trend_df["count"].rolling(window=3, center=True).mean()

#plot
plt.figure(figsize=(14, 6))
plt.plot(trend_df["year_month"], trend_df["rolling"], color="red")
plt.title("TSA Complaint Trend (3-Month Rolling Avg)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



Visual 5 — Category Totals Bar Chart

```
[14]: # import library
from matplotlib.ticker import ScalarFormatter

# define
cat_totals = (
    category_df.groupby("category")["count"]
    .sum()
    .reset_index()
)

# Filter out categories with extremely low totals
cat_totals = cat_totals[cat_totals["count"] >= 50]

# Sort descending
cat_totals = cat_totals.sort_values("count", ascending=False)

# Shorten long labels
cat_totals["short_label"] = cat_totals["category"].apply(
    lambda x: x[:40] + " ..." if len(x) > 40 else x
)

# plot
plt.figure(figsize=(12, 12))

# create horizontal lines
plt.hlines(
    y=cat_totals["short_label"],
    xmin=0,
    xmax=cat_totals["count"],
```

```

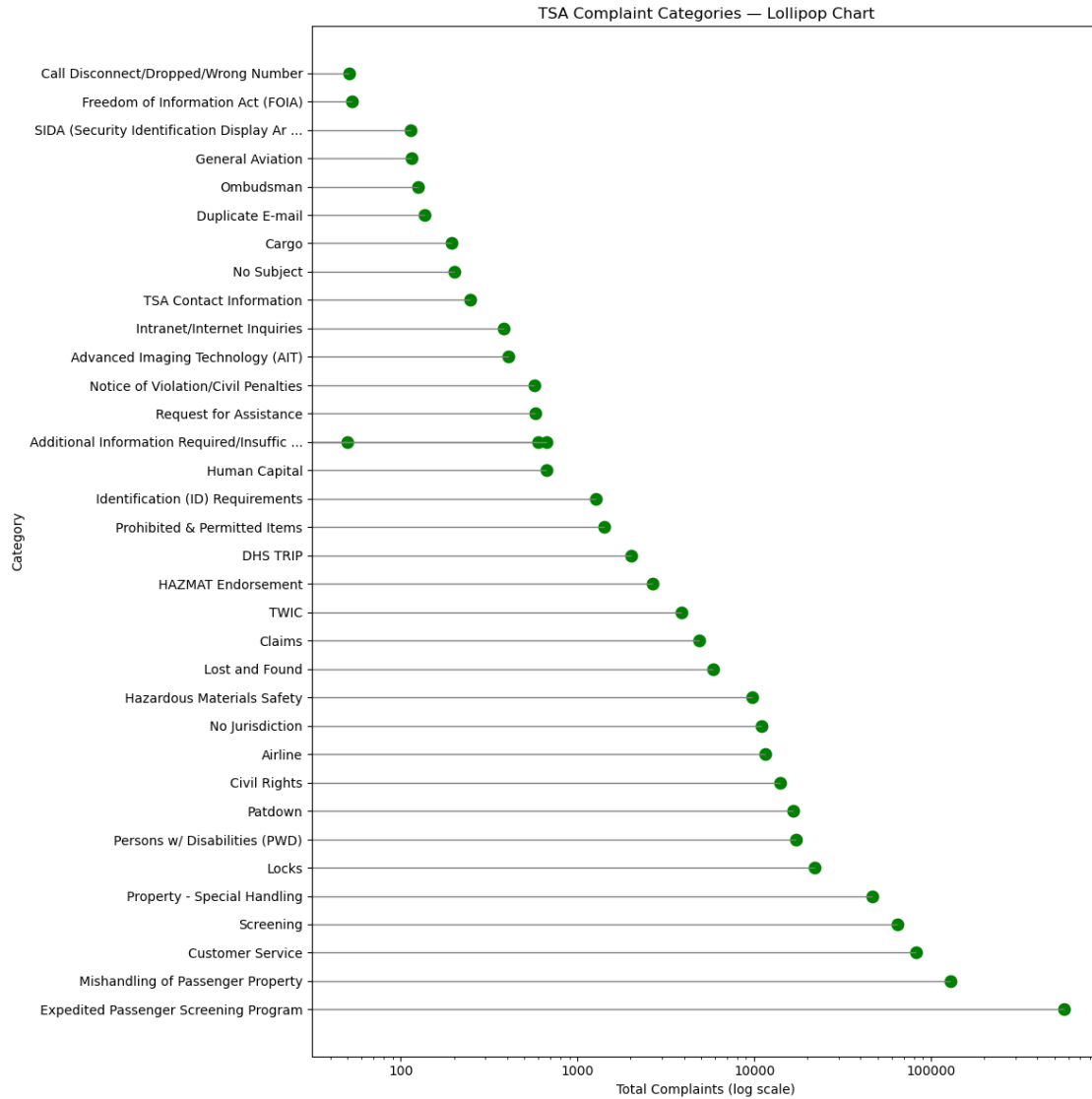
        color="gray",
        linewidth=1
    )

    # plant dots
    plt.scatter(
        cat_totals["count"],
        cat_totals["short_label"],
        color="green",
        s=80
    )

    # Log scale without scientific notation
    plt.xscale("log")
    ax = plt.gca()
    ax.xaxis.set_major_formatter(ScalarFormatter())
    ax.ticklabel_format(style='plain', axis='x')

    plt.xlabel("Total Complaints (log scale)")
    plt.ylabel("Category")
    plt.title("TSA Complaint Categories - Lollipop Chart")
    plt.tight_layout()
    plt.show()

```



Visual 6 — Horizontal Bar Chart (Top Subcategories)

```
[15]: # Aggregate subcategory totals
sub_totals = (
    subcategory_df.groupby("subcategory")["count"]
    .sum()
    .reset_index()
)

# Keep non-zero subcategories totals
sub_totals = sub_totals[sub_totals["count"] > 0]

# Sort to top 15 for readability
```

```

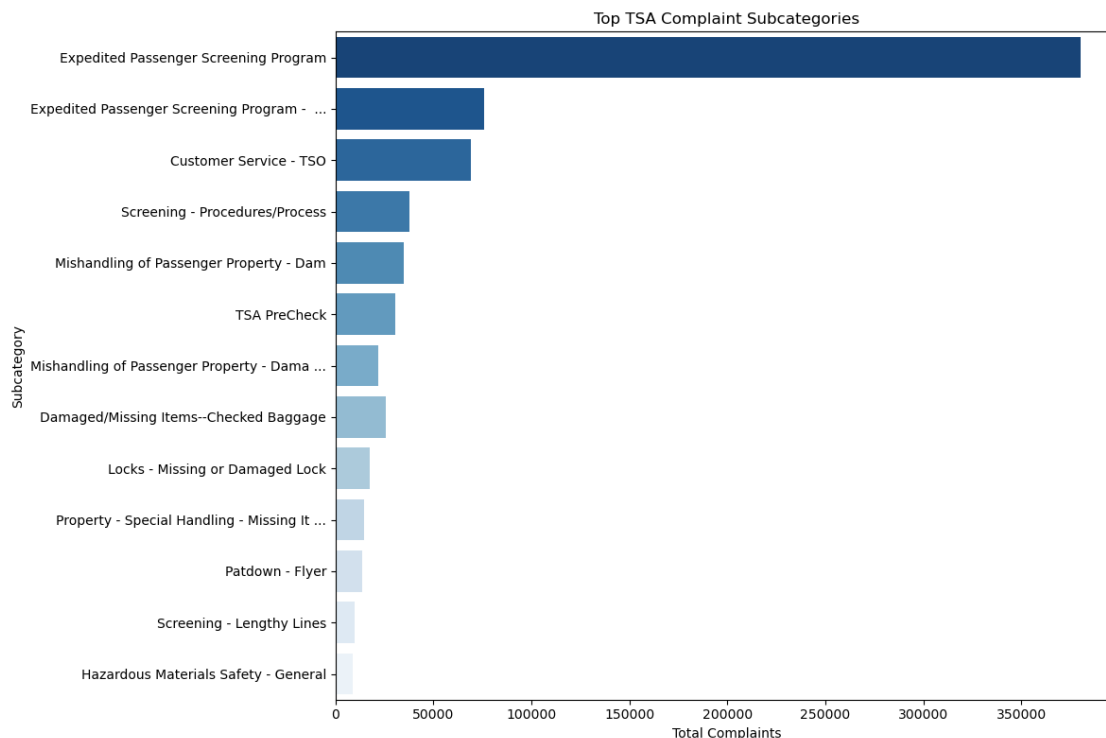
sub_totals = sub_totals.sort_values("count", ascending=False).head(15)

# Shorten labels for readability
sub_totals["short_label"] = sub_totals["subcategory"].apply(
    lambda x: x[:40] + " ..." if len(x) > 40 else x
)

# plot
plt.figure(figsize=(12, 8))
sns.barplot(
    data=sub_totals,
    x="count",
    y="short_label",
    hue="short_label",
    palette="Blues_r",
    errorbar=None,
    legend=False
)

plt.title("Top TSA Complaint Subcategories")
plt.xlabel("Total Complaints")
plt.ylabel("Subcategory")
plt.tight_layout()
plt.show()

```



Narrative Write-up I framed this story for an internal TSA operations audience, the kind of group that already works around the data but rarely sees it pulled together in a way that shows where the system is straining. These are supervisors, program managers, and field leads who understand the mechanics of screening and customer interaction but do not have time to dig through raw complaint files. They need a view that is direct, pattern-focused, and tied to decisions they can actually make.

The purpose of the story is to surface where complaint volume is consistently concentrated and what that means for staffing, training, and process control. The data shows that a small set of categories drives most of the friction. It also shows that the pressure points are not random. They repeat across airports and across time. The call to action is simple. Treat these categories as operational defects that can be reduced with targeted interventions instead of accepting them as background noise.

I chose a slide-based presentation because this audience is used to short operational briefings. They want to see the visuals, hear the implications, and move straight to next steps. A dashboard would have forced them to explore on their own. A written report would have buried the signal. The slide format lets the visuals carry the weight while the narrative stays tight.

The design choices follow the same logic. Color stays functional, not decorative. Red is used consistently to mark volume and intensity so the eye goes straight to the problem areas. Text stays minimal and direct. Alignment and spacing keep the visuals readable without forcing the viewer to hunt for labels. Log scales and capped color ranges are used only where the data requires it, and the intent is always to reveal structure rather than distort it.

The six visuals work together to show the system from different angles. The heat map shows how complaint categories behave over time. The box plot shows how unstable some categories are month to month. The spatial visual grounds the problem in physical space and makes it clear that a handful of airports carry disproportionate load. The time series trend shows the long arc of volume and the periods where the system was under stress. The lollipop chart makes the category hierarchy visible in a way a bar chart cannot. The subcategory bar chart shows exactly where the operational breakdowns occur.

The ethical considerations are straightforward. The data was used as provided, with only cleaning steps that preserved meaning. No categories were removed without explanation. Log scaling and percentile caps were applied to prevent a few extreme values from flattening the rest of the distribution. The main risk is misinterpretation, especially when using log scales or capped color ranges, so the visuals are labeled clearly and the narrative explains why those choices were made. The data comes from public TSA sources and does not contain personal information. The analysis stays focused on system performance, not individual behavior. The mitigation is transparency in how the data was shaped and why.